at the centers of supernova explosions. Since these strong-gravity regions are typically surrounded by thick layers of matter that absorb electromagnetic waves (but do not absorb gravitational waves), the strong-gravity regions cannot send us electromagnetic waves. The electromagnetic waves seen by astronomers come, by contrast, almost entirely from weak-gravity, low-velocity regions; for example, the surfaces of stars and supernovae.

These differences suggest that the objects whose symphonies we might study with gravitational-wave detectors will be largely invisible in light, radio waves, and X-rays; and the objects that astronomers now study in light, radio waves, and X-rays will be largely invisible in gravitational waves. The gravitational Universe should thus look extremely different from the electromagnetic Universe; from gravitational waves we should learn things that we will never learn electromagnetically. This is why gravitational waves are likely to revolutionize our understanding of the Universe.

It might be argued that our present electromagnetically based understanding of the Universe is so complete compared with the optically based understanding of the 1930s that a gravitational-wave revolution will be far less spectacular than was the radio-wave/X-ray revolution. This seems to me unlikely. I am painfully aware of our lack of understanding when I contemplate the sorry state of present estimates of the gravitational waves bathing the Earth. For each type of gravitational-wave source that has been thought about, with the exception of binary stars and their coalescences, either the strength of the source's waves for a given distance from Earth is uncertain by several factors of 10, or the rate of occurrence of that type of source (and thus also the distance to the nearest one) is uncertain by several factors of 10, or the very existence of the source is uncertain.

These uncertainties cause great frustration in the planning and design of gravitational-wave detectors. That is the downside. The upside is the fact that, if and when gravitational waves are ultimately detected and studied, we may be rewarded with major surprises.

In 1976 I had not yet become pessimistic about bar detectors. On the contrary, I was highly optimistic. The first generation of bar detectors had